

The sandwich

Run any model in laplace's coordinates, map back exactly. Everyone improves.

$z = \Phi^{-1}(F_t(y))$ the laplace transform

lettuce: a near-i.i.d. $N(0,1)$ stream, drift and volatility removed

any model: Prophet · ETS · AutoARIMA · GARCH · TabFM · PyMC

cheese: its z-space density, whatever shape it likes

$\log f_1(y) = \log f_2(z) + \log f_t(y) - \log \phi(z)$ exact inverse

Featured sandwich of the day: Prophet. Raw, it trails laplace by 0.76 nats median across 921 series (4.6 family-weighted). Sandwiched, 0.02. That is 97% of its density gap closed without retraining anything, calendar machinery intact, via [pip install prophet-laplace](#).

laplace defines a causal bijection on paths: the [Rosenblatt transform](#) of each observation under the predictive issued for it. Feed the resulting z stream to any model, map its output back through the exact Jacobian, and the model operates on a stationarised, unit-scale stream instead of raw data. No retraining, no approximation in the accounting. The [skill](#) has the recipe; [prophet-laplace](#) is the drop-in package for Prophet.

Forecasters, one-step log-likelihood vs plain laplace

model	universe	raw	sandwiched
ETS	FRED-30	-2.03	+0.01

model	universe	raw	sandwiched
AutoARIMA	FRED-30	−2.03	+0.05
GARCH(1,1)	FRED-30	−1.94	+0.03
Prophet	FRED-30	−2.04	+0.03
Prophet	921 series, pre-registered	−0.76 (−4.6 family-weighted)	−0.02
TabFM (regressor)	226 series	−0.75	−0.02
TabFM (x100 classifier)	921 series, pre-registered	−0.15	+0.87 family-weighted; +1.99 on repeat-heavy series
PYMC-Forecast (Student-t lag regression, ADVI)	226 series, pre-registered	−0.45	+0.04; +2.1 on repeat-heavy series

Numbers are median nats per observation relative to laplace alone (FRED-30 rows are means relative to laplace's 3.674). The pattern: raw models trail by whole nats; sandwiched, they converge to laplace plus or minus a small epsilon, and the epsilon measures conditional structure laplace missed. The large positive epsilons sit on repeat-heavy series, first found by the discrete TabFM head and confirmed by the PYMC-Forecast bout; that structure is a standing library work item.

The PYMC-Forecast row is the first sandwiched challenger whose positive epsilon spans the whole universe rather than one stratum: ahead of laplace on 65% of series, +0.04 nats median, filed as its pre-registered convergence hypothesis. The z stream also suits Bayesian machinery unusually well. Its marginal is $N(0,1)$ by construction, so priors centred at no structure and unit scale are exactly right rather than guesswork, and a single pre-test ADVI fit frozen through all 150 test steps still ties laplace to the third decimal: in these coordinates, batch-refit staleness costs almost nothing. One pooled hierarchical fit across all 226 series lost to the same model fit per series on 72% of them, with the gains concentrated on the repeat-heavy stratum (+2.1 nats median); the full tape is on the [bout page](#).

That convergence is not a daily-universe artifact. The same frozen z-space fit, carried onto the seasonal and waveform arms, pulls *ahead* of plain laplace rather than merely tying it: it wins 55% of the weekly series, 64% of the monthly, and 84% of the M4-Hourly waveforms,

a small median lift (+0.009, +0.017, +0.063 nats) that grows with periodic structure and peaks on the near-deterministic hard waveforms. Those residual streams still carry conditional structure the lag regression reads back, most of all where the signal is a repeating shape (see `benchmarks/pymc_arms_study.py`). Standalone on the raw series it stays a genuine underdog, losing to laplace on four of five regimes (the challengers [radar](#)); the lift is the sandwich, not the model.¹

Detectors

detector	measure	raw	sandwiched
DSPOT (EVT thresholding)	UCR accuracy, 250 series	0.120	0.232
DSPOT	alarm rate at nominal 1e-3 (FRED, median)	2.3e-3	2.5e-3, and best-in-table deep-tail behavior
RRCF (random cut forest)	UCR accuracy	0.244	0.236 (a wash; carries its own normalizer)
left-discord matrix profile	UCR accuracy, 150 series	0.587	0.427 (hurt; z whitens its templates)

The detector rows locate the boundary: the sandwich lifts methods whose theorems want a stationary, calibrated input, does little for weak structural methods, and hurts template matchers. It transfers the forecaster's competence, in both directions.

Full protocols, statements filed before results, and every table's source: the [challengers page](#), the paper's sandwich section, and `benchmarks/anomaly/RESULTS.md` in the repository.

¹ Both are flexible toolkits, not single models: skaters composes transforms, leaves, and ensembles, while PyMC-Forecast lets you specify arbitrary probabilistic models subject to Bayesian updating. These initial benchmarks use only each package's *default* settings for both. Submissions of other models and combinations are very welcome; open a [pull request](#) or an [issue](#).

Every study here is reproducible from the harnesses committed under `benchmarks/`, each beside the results it produced.